

Zehao Xiao

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Professional Summary

I am a Postdoctoral Researcher at Huawei Noah's Ark Lab in Paris, working on **time-series foundation-model pre-training, adaptation, and evaluation**. I am a research contributor to **Tabby**, an open pre-training recipe for time-series foundation models, and first author of **PaCTS**, a 2026 arXiv preprint on instance-adaptive prompts for frozen forecasters (earlier version: **ICML 2026 Foundation Models for Structured Data Workshop Oral**). My Ph.D. at the University of Amsterdam with Cees Snoek focused on **generalization under distribution shift** and **test-time adaptation**, with first-author papers at ICLR and CVPR. My research spans probabilistic inference, meta-learning, efficient foundation-model adaptation, and distributed PyTorch training.

Education

University of Amsterdam

Ph.D. in Computer Science and Machine Learning

Thesis: *Learning to Generalize at Test Time*

Supervisor: *prof. dr. Cees Snoek* (Head of VIS Lab, ELLIS Fellow, Scientific Director of Amsterdam AI, Netherlands Prize for ICT Research)

Amsterdam, the Netherlands

June. 2020 - June. 2025

Beihang University

Master in Information and Communication Engineering

MS project: *Crowd Counting and Density Estimation*

Beijing, China

Sep. 2017 - Feb. 2020

Beihang University

Bachelor in Electronic Engineering

Beijing, China

Sep. 2013 - Jun. 2017

Publications

- [1] **Z Xiao**, S Xie, L Zan, J Zhang, L Pan, I Redko, M Tiomoko, K Zhang. "Instance-Adaptive Prompts as Context for Time-Series Foundation Models," [\[arXiv preprint 2026\]](#) (Earlier version: ICML 2026 FMSD Workshop Oral)
- [2] **Z Xiao**, S Yan, J Hong, J Cai, X Jiang, Y Hu, J Shen, Q Wang, C Snoek. "DynaPrompt: Dynamic Test-Time Prompt Tuning," [\[ICLR 2025\]](#)
- [3] **Z Xiao**, J Shen, M Derakhshani, S Liao, C Snoek. "Any-Shift Prompting for Generalization over Distributions," [\[CVPR 2024\]](#)
- [4] **Z Xiao**, C Snoek. "Beyond Model Adaptation at Test Time: A Survey," [\[Preprint 2024\]](#)
- [5] **Z Xiao**, X Zhen, S Liao, C Snoek. "Energy-Based Test Sample Adaptation for Domain Generalization," [\[ICLR 2023\]](#)
- [6] **Z Xiao**, X Zhen, L Shao, C Snoek. "Learning to Generalize across Domains on Single Test Samples," [\[ICLR 2022\]](#)
- [7] **Z Xiao***, J Shen*, X Zhen, L Shao, C Snoek. "A Bit More Bayesian: Domain-Invariant Learning with Uncertainty," [\[ICML 2021\]](#) (Equal contribution)
- [8] S Xie, B Abdesslem, **Z Xiao**, Y Attia El Hili, A Odonnat, Z Dong, L Zan, T Palpanas, J Zhang, L Pan, K Zhang, M Tiomoko. "Tabby: An Open Pretraining Recipe for Time Series Foundation Models," [\[Preprint \(Technical Report\) 2026\]](#) (Core research contributor)
- [9] S Deng, **Z Xiao**, M de Rijke. "Adaptive Latent Decomposition for Domain Generalization in Time Series Forecasting," [\[ACM Transactions on Knowledge Discovery from Data 2026\]](#)
- [10] C Wang*, **Z Xiao***, Y Mao*, Y Qu*, J Shen, Y Lv, X Ji. "Model Predictive Task Sampling for Efficient and Robust Adaptation," [\[Nature Communications 2026\]](#) (Equal contribution)
- [11] W Yin*, **Z Xiao***, P Zhou, S Yu, J Shen, J Sonke, E Gavves. "Distributional Vision-Language Alignment by Cauchy-Schwarz Divergence," [\[ICLR 2026\]](#) (Equal contribution)
- [12] S Ambekar*, **Z Xiao***, J Shen, X Zhen, C Snoek. "Probabilistic Test-Time Generalization by Variational Neighbor-Labeling," [\[CoLLAs 2024\]](#) (Equal contribution)
- [13] X Jiang*, **Z Xiao***, B Zhang, X Zhen, X Cao, D Doermann, L Shao. "Crowd Counting and Density Estimation by Trellis Encoder-Decoder Networks," [\[CVPR 2019\]](#) (Equal contribution)
- [14] J Shen, Q Wang, **Z Xiao***, N van Noord, M Worring. "GO4Align: Group Optimization for Multi-Task Alignment," [\[NeurIPS 2024\]](#) (Corresponding author)
- [15] W Yin, P Zhou, **Z Xiao**, J Liu, S Yu, J Sonke, E Gavves. "Towards Uniformity and Alignment for Multimodal Representation Learning," [\[ICML 2026\]](#)
- [16] W Yin, **Z Xiao**, J Shen, Y Chen, C Snoek, J Sonke, E Gavves. "Geometric Neural Process Fields," [\[TMLR 2026\]](#)
- [17] J Liu, P Zhou, **Z Xiao**, J Shen, W Yin, J Sonke, E Gavves. "Probabilistic Interactive 3D Segmentation with Hierarchical Neural Processes," [\[ICML 2025\]](#)
- [18] S Ambekar, **Z Xiao**, J Shen, X Zhen, C Snoek. "GeneralizeFormer: Layer-Adaptive Model Generation Across Test-Time Distribution Shifts," [\[WACV 2025\]](#)
- [19] Y Du, **Z Xiao**, S Liao, C Snoek. "ProtoDiff: Learning to Learn Prototypical Networks by Task-Guided Diffusion," [\[NeurIPS 2023\]](#)
- [20] J Shen, **Z Xiao**, X Zhen, C Snoek, M Worring. "Association Graph Learning for Multi-Task Classification with Category Shifts," [\[NeurIPS 2022\]](#)

Research Experience

Foundation models for time series and forecasting

Huawei Paris Noah’s Ark Lab2026

Research on **time-series foundation models** across pre-training, efficient post-training, and evaluation. Contributed to **Tabby**, an open pre-training recipe and 145M-parameter probabilistic forecaster (technical report, 2026), using **FSDP/DDP multi-GPU training**. Led **PaCTS**, a first-author 2026 preprint that conditions a frozen forecaster on compact instance-adaptive prompts; an earlier version was an **ICML workshop Oral**.

Generalization and adaptation across data distributions

University of Amsterdam2020 - 2026

Developed **test-time adaptation** methods for pretrained models under distribution shift: online prompt tuning for vision-language models (**ICLR 2025**), prompt-based generalization across distributions (**CVPR 2024**), single-sample variational adaptation (**ICLR 2022**), and energy-based test-sample adaptation (**ICLR 2023**). Authored a survey of test-time adaptation and supervised a Master’s thesis that led to papers at CoLLAs 2024 and WACV 2025.

Meta-learning and efficient task selection

University of Amsterdam2022 - 2026

Co-developed **model-predictive task sampling** to estimate adaptation risk before training and select tasks for efficient, robust learning (**Nature Communications 2026**, equal contribution). Contributed to **ProtoDiff**, which uses task-guided diffusion to generate representations for few-shot learning (**NeurIPS 2023**).

Representation learning and transfer across tasks

University of Amsterdam2022 - 2026

Studied **distributional vision-language alignment** (**ICLR 2026**, equal contribution) and multimodal representation learning (**ICML 2026**); contributed to multi-task learning methods for category shifts and task imbalance (**NeurIPS 2022/2024**).

Probabilistic modeling and neural processes

University of Amsterdam2024 - 2026

Contributed to neural-process models for **interactive 3D segmentation** (**ICML 2025**) and geometric neural fields (**TMLR 2026**), with uncertainty-aware predictions and evaluation.

Computer vision and crowd density estimation

Beihang University2018 - 2019

Researched convolutional and attention-based models for crowd counting and density estimation, contributing to publications at **BMVC 2018**, **CVPR 2019**, and **ICCV 2019**.

Teaching

Teaching assistant for coursesComputer Vision (2021, 2022), Deep Learning (2022), Foundation Models (2024)

Supervisor on Master’s thesisSameer Ambekar. “Learning to Generate Labels and Models at Test-Time” (2023)

Technical Skills

ProgrammingPython, PyTorch, TensorFlow

Drawing & TypesettingL^AT_EX, Office

LanguagesEnglish, Chinese(Native)

Volunteer Services

Conference Reviewer

NeurIPS, ICML, ICLR, CVPR, ICCV, ECCV, AAAI

Journal Reviewer

TPAMI, IJCV, TNNLS